



Cultural Participation and Life Satisfaction: Is the Engagement in the Arts a Benefit for Everyone?

Romain Lerouge¹ · Juan Prieto-Rodriguez²

Received: 13 October 2025 / Accepted: 12 July 2026
© The Author(s) 2026

Abstract

Sustaining and preserving well-being has become a critical priority for policymakers, and a primary goal for institutions such as the European Union. In this context, renewed attention has been directed towards the role of cultural participation, deemed to be a crucial trigger of socio-economic development and social inclusion. Several studies have explored the relationship between cultural participation and well-being, often measured through life satisfaction. Results highlight a positive, but modest, association. Specifically, while the effects on life satisfaction are analysed at an individual level, they are the results of more complex intergenerational dynamics. In fact, taste formation and habits are greatly affected by parents through education and imitation mechanisms. These, in turn, affect not only the propensity to participate, but also the ability to derive utility from cultural participation. This paper employs mixture models (FMM) to examine the association between cultural participation and life satisfaction, assessing how it is moderated by family background. Our findings show a positive link, especially for disadvantaged social groups, where family background sometimes acts as a barrier to fully benefiting from engagement in the arts.

Keywords Cultural participation · Life satisfaction · Family background · Well-being

1 Introduction

In 2007, in the Treaty of Lisbon, the European Union amended its primary aim ‘*to promote peace, its values and the well-being of its people*’ (European Union, 2007). This act manifested the centrality of well-being in Western societies and the growing attention towards this aspect for public agencies. The importance of well-being is explicitly enshrined in numer-

✉ Romain Lerouge
romain.lerouge@polimi.it

Juan Prieto-Rodriguez
juanprieto@uniovi.es

¹ Politecnico di Milano, Milano, Italy

² Universidad de Oviedo, Oviedo, Spain

ous academic studies (Blanchflower & Clark, 2021; Han & Kaiser, 2024) and other statements promoted by governments and institutions such as the United Nations (2023), OECD (2013) or UNESCO (2022), aiming at promoting a more comprehensive spectrum of social, economic and environmental benefits. Despite there is no universal definition, well-being entails a comprehensive evaluation of people's lives, encompassing *'good mental states, including all of the various evaluations, positive and negative, that people make of their lives and the affective reactions of people to their experiences'* (OECD, 2013) and it is deemed as a complement to the traditional economic measures of welfare such as GDP per capita.

In this perspective, renovated attention is emerging towards the role of culture as a factor strictly related to well-being. *'Far from being isolated from ordinary experience, the arts, through their communicative power, enhance individual engagement with the world in ways that have both personal and public benefits'* (McCarthy et al., 2004). This interest has catalysed in particular on cultural participation, considered to be a trigger of socio-economic change (Cerisola & Panzera, 2022) and to be able to enhance people's enrichment, democratisation and social inclusion (Stevenson et al., 2017). Cultural participation includes a broad set of activities deliberately carried out by a person conducting cultural value and symbolic attributes (Pessoa & Deloumeaux, 2009), and they can be categorised into active and passive activities (van Wel et al., 2006; Lee et al., 2021). Although the distinction between the two may at times be ambiguous (Węziak-Białowska, 2016), active participation entails direct engagement in the creation of artistic expression, whereas passive participation refers to attendance at a cultural event or experience.

In recent years, a growing body of research has explored the relationship between cultural participation and well-being. While this latter concept is captured through various measures, including perceived health status (Fancourt et al., 2026; Jensen et al., 2023; Lindström et al., 2024), most research concentrates more specifically on its cognitive long-term component, namely life satisfaction. Life satisfaction is defined as the self-evaluation of a person's life (Pavot & Diener, 1993). This metric is emerging as a reliable non-market evaluation tool that may help to quantify the positive externalities generated by cultural engagement, as it is presumed to remain consistent over time and unaffected by temporary fluctuations in mood (Diener, 2009). In other words, an accurate assessment of life satisfaction facilitates the evaluation of the relationship between cultural participation and individual utility (Baldin & Bille, 2023).

Building on this body of evidence, several large-scale longitudinal studies have examined the pathways through which cultural and arts engagement may improve well-being outcomes. Drawing on extensive review evidence, Fancourt and Finn (2019) document that arts engagement is associated with reduced levels of depression and anxiety, as well as improvements in subjective well-being and quality of life. Fancourt and Steptoe (2019), using data from the English Longitudinal Study of Ageing, find that attendance at cultural events is associated with a lower risk of mortality, with effects mediated partly through the psychosocial and emotional pathways activated by engagement.

As a result, much of the recent literature analysing the effect of the consumption of cultural goods, has reported a positive association with life satisfaction (Baldin & Bille, 2023; Reyes-Martínez et al., 2021). For instance, some studies have identified beneficial effects from attending theatre performances (Baldin & Bille, 2023; Viola et al., 2024), visiting museums (Lee et al., 2021; Vegheş, 2020; Viola et al., 2024), and cinema and libraries attendance (Vegheş, 2020). In this strain of the literature, primarily due to the lack of infor-

mation, active cultural participation is often underemphasized and is frequently measured through aggregated constructs (Grossi et al., 2012; Iida, 2019; Nenonen et al., 2014). In fact, active participation is often omitted from the analysis altogether (Baldin & Bille, 2023; Lee et al., 2021; Reyes-Martínez et al., 2021; Viola et al., 2024). However, a few studies have found a positive association between life satisfaction of active cultural practices such as dancing (Brown et al., 2015) as well as painting and music-related experiences such as singing or playing an instrument (Iida, 2019).

However, other researchers have contested these outcomes, reporting a weak (Michalos & Kahlke, 2008), or even non-existent relationship between cultural participation and life satisfaction (Węziak-Białowolska, 2016). Moreover, Reyes-Martínez (2021), analysing different ethnic groups in Latin America, identified effects only among certain population segments. The dynamics underpinning the link between cultural engagement and life satisfaction are thus still unclear (Lee et al., 2021) and lack a proper formalized framework that accounts for the potential heterogeneity of this relationship characterised by persistent social inequalities (Fancourt & Warran, 2024; Reyes-Martínez et al., 2021). To address this gap, we analyse family background as a crucial factor in moderating and explaining the potential association between cultural participation and life satisfaction, particularly in the context of complex intergenerational dynamics.

Through education and imitation mechanisms, the family plays a crucial role in the transmission of values and behaviours. This aspect is manifested through both an indirect and direct effect. First, family influences personal taste and predisposition to engage in cultural activities (Klokke & Jæger, 2022; Suarez-Fernandez et al., 2020), which in turn can affect life satisfaction.

Many academics, building on the seminal work of Bourdieu (1984), have indeed argued that cultural participation serves as a means of social distinction, mainly accessed by advantaged social groups. Membership in a particular social group influences individuals' likelihood of engaging in cultural activities, which in turn tends to benefit more those participants with higher social capital, thereby exacerbating, rather than mitigating, social inequalities (Klokke & Jæger, 2022; Mak et al., 2021; Hanquinet et al., 2019). The influence of social background on cultural participation is described and extended in the omnivore theory, which states that elites embrace a larger and more diverse spectrum of cultural activities, unlike other social groups, who exhibit more limited and specialised interests (Peterson, 1992; Peterson & Kern, 1996).

In addition to this first effect, family background directly affects people's attitudes and behaviours. This translates into differences in people's ability to extract utility from cultural participation, and benefit from higher levels of life satisfaction in general. While existing research mainly discusses the influence of children on parents, that we could describe as a form of "reverse heritage influence" (Blanchflower & Clark, 2021), few studies also point out the hereditary transmission of well-being has been observed in several studies, revealing a strong influence (Schnitzlein & Wunder, 2016; Headey et al., 2014), across a large spectrum of countries (Molina et al., 2011). This phenomenon has been theorised and modelled by the equality of opportunity theory (Roemer, 1993), which states that the result of people's actions is in part explained by factors lying behind personal responsibility, and most notably by family background (Bourguignon et al., 2007). In other words, this theory posits that to evaluate the result of people's actions one's must isolate it from the effect explained by those exogenous factors. The analysis of those imbalances is especially relevant for policy-

makers, who are more interested in addressing inequalities explained by those exogenous factors, rather than the overall unbalance in life satisfaction (Checchi & Peragine, 2010).

The role of family background is thus crucial in shaping both individuals' preferences and choices and their ability to extract utility from these choices, including cultural consumption decisions. However, in extant literature its role has been overlooked, implicitly assuming that the benefits of cultural participation on life satisfaction are uniform across all individuals. Moreover, most studies focus on a limited set of activities, neglecting a comprehensive overview of cultural expressions and hiding existing patterns of cultural omnivorousness and the role of active cultural participation, despite its unique characteristic whereby individuals are directly involved in the generation of the cultural form, contributing to a bottom-up generation of value. Considering these gaps in the existing literature, this paper investigates the relationship between cultural participation and life satisfaction, assessing how it is moderated by family background.

Existing studies assume implicitly that the utility function mapping cultural participation onto life satisfaction is homogeneous across the population — that is, the same activity generates a similar average benefit for everyone. This assumption contrasts with the heterogeneous returns to cultural capital documented by Bourdieu (1984), and with the observed fragility of aggregate estimates across different population groups (Węziak-Białowolska, 2016; Michalos & Kahlke, 2008). We address this gap directly by employing Finite Mixture Models (FMM), which allow us to identify, endogenously, population subgroups with structurally different relationships between cultural participation and life satisfaction. The use of FMM moves the analysis beyond the question of whether cultural participation matters on average, and towards the question of for whom it matters and through which mechanisms.

Cultural participation is defined as the engagement in a large array of active and passive cultural activities. Moreover, to account for the existence of different behavioural and attitudinal factors potentially affecting the relationship with life satisfaction, the study utilizes finite mixture models (FMM), which are capable of addressing the large unobserved heterogeneity between sub-groups in the population.

The paper is organised as follows. The next section illustrates the database that has been developed and elaborated for this study. Section 3 explains the empirical strategy adopted to pursue the research objective. Then, Sect. 4 presents and discusses the results of the study. The final section summarises the main conclusions of the research.

2 Data

To conduct the research, a survey was sent out in March 2023 to a sample of people across Italy, representative of the population with access to the internet in terms of age, gender, education and geographical area. Italy serves as a perfect case study due to its rich and diverse cultural landscape. Indeed, it has previously been chosen as the empirical setting in several studies (Grossi et al., 2012; Viola et al., 2024), while also being characterized by nation-wide social imbalance and inequality of opportunity (Checchi & Peragine, 2010), including those directly related to life satisfaction intergenerational transmission (Molina et al., 2011). The survey was disseminated by an external provider using Computer Assisted Telephone Interviewing (CATI) methodology. The final sample comprises 1,286 questionnaires (retention rate = 85.7%), with their variables described in the following paragraphs.

To assess *cultural participation*, respondents were asked about their participation over the last year in 12 different cultural activities, 6 actives and 6 passives, typically included while studying cultural participation (Reyes-Martinez, 2021; van Wel et al., 2006). Passive cultural participation encompasses visits to museum exhibitions and archaeological sites; theatre; cinema; libraries; concerts and reading books. Active cultural participation includes playing an instrument; singing; dancing; drawing, painting and sculpting; writing and poetry; photography and videomaking and, finally, acting. Moreover, participation in events was reported on a four-level scale: “Never”, “Once a year”, “Few times a year”, “Regularly”. Instead, individual activities encompass three levels of intensity: “Never”, “Sometimes”, “Regularly”.

Life satisfaction (LS) is measured using the Satisfaction with Life Scale (SWLS), developed and validated by Pavot and Diener (1993). This is a widely used measure in the existing literature due to its consistency over time, independence from temporary emotional states, and its capacity to capture multiple aspects of people’s lives (Diener, 2009). It is thus considered a good proxy of individual utility (Baldin & Bille, 2023). The metric consists of five questions, to which the respondent answers on a 7-point Likert scale. The final life satisfaction score is obtained by summing the resulting scores of the five questions, thus obtaining a measure ranging from 0 to 30.

Life satisfaction is measured as the sum of the five SWLS items (developed and validated by Pavot & Diener, 1993), yielding a scale from 0 to 30. Summing Likert items into a composite score implicitly assumes cardinality — equal psychological distances between scale points — and permits treating the outcome as continuous in the FMM estimation. This assumption is standard in the life satisfaction literature (Diener, 2009; Baldin & Bille, 2023). Overall, this measure is widely used in the existing literature due to its consistency over time, independence from temporary emotional states, and its capacity to capture multiple aspects of people’s lives (Diener, 2009).

Family background is assessed by the highest level of education attained by the parents. This proxy is widely used in the existing literature (Klokke & Jæger, 2022; Reeves & de Vries, 2019) and frequently serves to quantify inequalities of opportunity (Perez-Mayo, 2019; Checchi & Peragine, 2010). This conceptual approach focuses on estimating parents’ level of human and social capital, relevant to describe the phenomenon of social distinction theorised by Bourdieu (1984). Additionally, following the methodology proposed by Báez-Montenegro et al. (2011), the study identifies the relative parent background adjusting it by the cohort of the respondent. This procedure accounts for the changing average degree of education in the different age cohorts (shown in Fig. 1). The Italian educational landscape has undergone significant changes over the past 50 years. The definition of ‘highly educated parents’ for individuals in their 70s differs substantially from that of individuals in their 20s today. Among the oldest participants in our sample, parents with a complete secondary education or higher were regarded as “educated”. However, for younger cohorts, this level of education is no longer sufficient, as their parents, on average, have nearly attained tertiary education on average. Consequently, a fixed absolute threshold, constant over time, is inadequate for classifying households by parents’ educational attainments. Instead, we adopt a relative criterion: individuals whose parents’ education is equal to or above the median level of education for their cohort are classified as coming from an advantaged background (FB=1). Those whose parents’ education falls below this threshold are identified as part of the most disadvantaged population in terms of family background (FB=0).

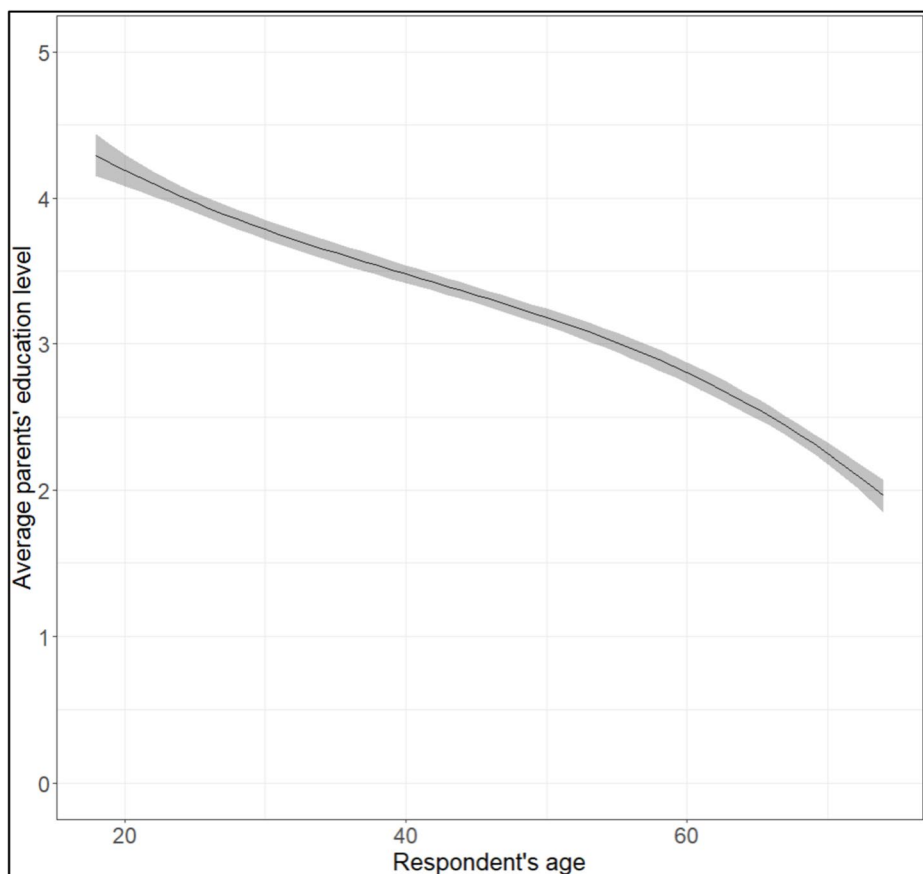


Fig. 1 Parents' average education level through age cohorts

Moreover, the study incorporates a set of control variables. *Educational attainment* is categorized into 6 levels in the survey, ranging from primary school to postgraduate education. In order to account for education as a continuous variable in the empirical models, these categories were converted into *years of education*, based on the standard duration of each educational level in Italy: 5 years for primary school, 3 for secondary school, 5 for high school, 3 for Bachelors degree, 2 for Master's degree and 2 for post-graduate. Education is a crucial determinant in shaping individuals' attitudes and aesthetic preferences, particularly influencing their propensity to engage with highbrow culture consumption (Suarez-Fernandez et al., 2020).

Then, annual *income* is included, measured on a 7-level scale from "no income" to "more than 100,000 euros/year". A total of 127 respondents preferred not to answer this question. Similarly to educational attainment, income was converted into a continuous variable by assigning each bracket its average value. Other control variables include *age* (from 18 to 75 years), *gender* (male or female), *geographical area* (North-west, North-East, Centre, or South and Islands), *city of residence size* (measured by the logarithmic of the number of inhabitants), *labour status* (worker, student, unemployed or retired), *sport activity* (yes or

no) and *religious faith* (believer or atheist/agnostic). The main descriptive statistics of all the survey variables are reported in Table 1.

To highlight the relationships among cultural participation activities, Fig. 2 presents the correlation matrix. This representation identifies patterns of cultural omnivorousness within the sample. Passive activities are positively correlated with each other, particularly theatre attendance and museum visits (+0.47), as well as cinema and concert attendance (+0.44). Hence, these passive activities tend to cluster into two distinct categories: popular culture and highbrow culture, with the highest correlations observed within each group. In contrast, correlations between activities across these two groups are relatively lower. On the other hand, active cultural activities exhibit weaker correlations among them, with the notable exception of singing and playing a musical instrument (+0.24).

These statistically significant correlations observed among specific cultural activities will be exploited to derive different profiles of cultural consumers. These profiles will subsequently serve as explanatory variables within behavioural models, which will be estimated through the application of Finite Mixture Models technique. The next section illustrates the empirical strategy pursued in the study to define such behaviour models.

Table 1 Descriptive statistics

Statistic	Mean	St. Dev.	Min	Max
Museum	0.746	0.899	0	3
Theatre	0.663	0.896	0	3
Cinema	0.863	1.024	0	3
Library	0.333	0.700	0	3
Concerts	0.526	0.768	0	3
Instrument	0.294	0.601	0	2
Reading	1.018	0.800	0	2
Sing	0.186	0.474	0	2
Dance	0.302	0.539	0	2
Draw	0.194	0.475	0	2
Writing	0.096	0.346	0	2
Acting	0.083	0.333	0	2
Photo & video	0.495	0.594	0	2
Life satisfaction	22.003	4.863	0	30
Non-disadv. Background	0.726	0.446	0	1
Years of education	13.78	5.551	5	20
Income	3.271	1.431	1	7
Age	47.170	16.780	18	74
Gender_male	0.484	0.5000	0	1
Sport activity	0.656	0.475	0	1
City population (log)	10.364	1.432	6.009	14.827
Religion_atheist	0.248	0.432	0	1
Labor status_unemployed	0.122	0.328	0	1
Labor status_worker	0.634	0.482	0	1
Labor status_retired	0.170	0.375	0	1
Region_north-east	0.190	0.392	0	1
Region_north-west	0.267	0.442	0	1
Region_centre	0.204	0.403	0	1
Region_south	0.340	0.474	0	1

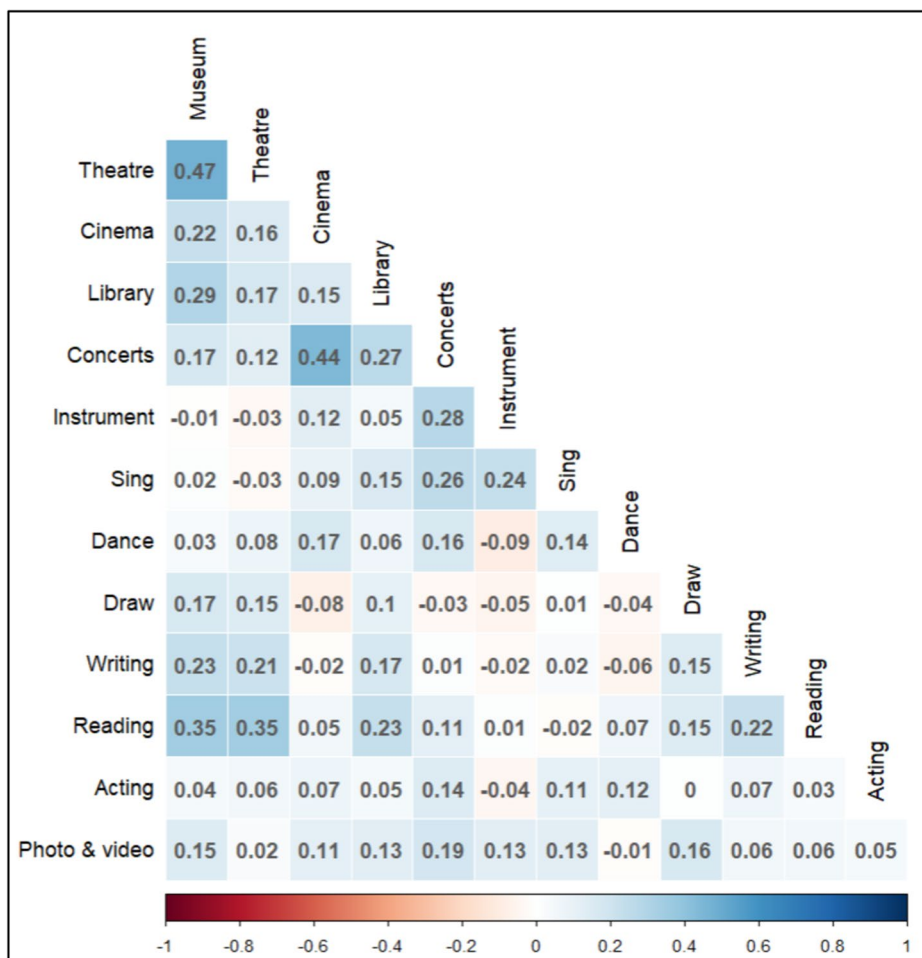


Fig. 2 Correlation matrix for cultural participation variables

3 Empirical Model

The objective of the study is to investigate the association between cultural participation and life satisfaction, with a focus on how this relationship is moderated by family background. In the existing literature, there is a common assumption that individuals share a similar utility function, which transforms consumption into life satisfaction. However, this assumption often fails to account for the direct and indirect influence exerted by family background. Taste formation is indeed transmitted by family both formally, through education, and informally, through imitation mechanisms.

To account for and identify different behavioural patterns of utility (i.e. life satisfaction) acquisition, we propose the use of Finite Mixture Models (FMMs) (McLachlan et al., 2019; Aitkin & Rubin, 1985). These models are particularly well-suited to account for unobserved heterogeneity (Arcidiacono & Jones, 2003; Kasahara & Shimotsu, 2009) and have been

used in analysing the cultural sector (Ateca-Amestoy, 2008; Fernandez-Blanco et al., 2009; Prieto-Rodriguez & Vecco, 2021; Vecco et al., 2024) and well-being (Ateca-Amestoy & Ugidos, 2013). The underlying assumption of this approach is that population heterogeneity arises from the presence of two or more distinct, yet unobserved, homogeneous subgroups, allowing for the existence of different behaviours across subsamples. These unobserved groups are determined endogenously through a classification equation. Hence, FMMs are characterised by two components that are estimated simultaneously.

The first part consists of modelling the behavioural equation—in this case, for each unobserved sub-group, a function that describes how they transform cultural consumption into higher life satisfaction. The second equation, instead, classifies the population into these different subgroups based on the distribution of the dependent variable and on a set of individual characteristics. These variables are known as “concomitants” (Masyn, 2013), and affect a person’s probability of belonging to each identified class or subgroup of the population. Thus, through FMMs, respondents are probabilistically separated into subgroups, each one with its own behavioural equation, allowing to describe how the link between cultural participation and life satisfaction changes across these subgroups. The definition of the two equations constituting the FMM is detailed in the following paragraphs.

We define the behavioural equation using life satisfaction as the dependent variable (y_i). Within each group—defined endogenously through finite mixture models—the determinants of life satisfaction include cultural participation (CP_i), the central variable of interest of this study. Specifically, we account for people’s engagement to a wide array of cultural activities, both passive and active, as introduced and described in “Data”, and group them into distinct profiles of cultural consumption. Moreover, the model also incorporates information about respondents’ sports activity (SP_i), an additional endeavour related to life satisfaction. Finally, personal income (INC_i) is included to account for the effect of hedonic purchases, which are also related to life satisfaction (Gladstone et al., 2024).

The behavioural equation is restricted to variables that represent individual choices or behaviours: cultural participation profiles, sports activity, and income. This restriction follows directly from the equality of opportunity framework (Roemer, 1993), which distinguishes between variables reflecting personal effort and decisions—for which individuals bear responsibility—and variables representing exogenous circumstances, primarily family background, which lie beyond individual control. Including family background in the behavioural equation alongside chosen activities would conflate the direct returns to voluntary engagement with a pre-existing baseline advantage that precedes any individual decision. Family background therefore enters the model exclusively as a determinant in the classification equation, where it captures the a priori probability of belonging to each latent class. This specification captures how exogenous social conditions shape the type of utility function an individual exhibits, without directly attributing life satisfaction to inherited circumstances. This approach is consistent with the equality of opportunity literature applied to well-being (Schnitzlein & Wunder, 2016; Molina et al., 2011) and preserves the interpretive integrity of the behavioural parameters as returns to chosen activities within each identified class.

$$y_i = f(x_i) = f(CP_i, SP_i, INC_i)$$

Moreover, the main hypothesis of this study is that this behavioural function could vary across distinct latent groups in the population. FMMs are used to identify groups with statis-

tically distinct mechanisms for transforming cultural participation into life satisfaction. The factors considered as potential determinants of the *a priori probability* of belonging to each of these groups, or classes, are age, gender, geographical area of living, city of residence's size, labour status, religious faith, education, and family background.

Considering this empirical strategy, the following section presents the main results of the study.

4 Results

At first, we want to check whether both life satisfaction and cultural participation vary with the family background. Table 2 reports the mean and the standard deviation of life satisfaction and cultural participation across disadvantaged and non-disadvantaged families, and the resulting t-test comparing whether these means are different. Results confirm the existence of a strong link between family background and life satisfaction, in line with existing research on the subject (Schnitzlein & Wunder, 2016; Headey et al., 2014). Moreover, the table shows that family background appears to be related also to the propensity to participate in some of the cultural activities included in this study, both active and passive. Specifically, people from disadvantaged families tend to practice less reading, drawing and photography, rarely play musical instruments, and attend museums, theatres and libraries fewer times. This confirms the high cultural omnivorousness of people coming from the most advantaged family backgrounds, who have a higher engagement in the majority of cultural typologies considered (Peterson & Kern, 1996). While these descriptive results do not uncover the mechanisms underlying the relationship between family background and either life satisfaction or cultural participation, they suggest that cultural engagement may serve as a pathway through which family background relates to life satisfaction.

The positive link between cultural participation and life satisfaction also supports the existence of this dynamic. Table 3 compares people's cultural participation within the first quantile of life satisfaction versus the others through univariate t-tests. Results show a significant

Table 2 Mean and Standard deviation (in parenthesis) by family background

	Disadvantaged Family		Non-disadvantaged family		P-value*
Live satisfaction	20.27	(4.925)	22.606	(4.575)	<0.001
INSTRUMENT	0.174	(0.379)	0.244	(0.430)	0.011
ACTING	0.055	(0.228)	0.073	(0.261)	0.268
DANCE	0.309	(0.463)	0.260	(0.439)	0.098
DRAW	0.129	(0.335)	0.177	(0.382)	0.048
SING	0.183	(0.388)	0.142	(0.349)	0.081
WRITING	0.016	(0.126)	0.104	(0.305)	<0.001
PHOTO_VIDEO	0.370	(0.484)	0.460	(0.499)	0.006
MUSEUM	0.322	(0.468)	0.535	(0.499)	<0.001
THEATRE	0.318	(0.467)	0.458	(0.499)	<0.001
CINEMA	0.502	(0.501)	0.483	(0.500)	0.572
LIBRARY	0.154	(0.362)	0.237	(0.426)	0.002
CONCERTS	0.373	(0.484)	0.390	(0.488)	0.607
READING	0.585	(0.493)	0.747	(0.435)	<0.001
N	311	(0.269)	847	(0.731)	

* P-value of the corresponding t-test on mean differences

Table 3 Mean and Standard deviation (in parentheses) by life satisfaction quartile

	Below the 1 st quartile of LS		1 st quartile of LS		P-value*
INSTRUMENT	0.229	(0.420)	0.219	(0.414)	0.706
ACTING	0.058	(0.234)	0.090	(0.287)	0.044
DANCE	0.280	(0.449)	0.257	(0.437)	0.405
DRAW	0.131	(0.338)	0.235	(0.425)	<0.001
SING	0.164	(0.371)	0.128	(0.335)	0.116
WRITING	0.052	(0.222)	0.089	(0.285)	0.025
PHOTO_VIDEO	0.413	(0.493)	0.486	(0.500)	0.019
MUSEUM	0.405	(0.491)	0.634	(0.482)	<0.001
THEATRE	0.342	(0.475)	0.590	(0.492)	<0.001
CINEMA	0.508	(0.500)	0.445	(0.498)	0.049
LIBRARY	0.179	(0.384)	0.292	(0.455)	<0.001
CONCERTS	0.383	(0.486)	0.391	(0.489)	0.792
READING	0.654	(0.476)	0.811	(0.392)	<0.001
N	311	(0.269)	847	(0.731)	

* P-value of the corresponding t-test on mean differences

difference in most of active and passive activities considered in the study (acting, drawing, photography and video-making, reading and attendance to museums, theatre, cinema and libraries) when people are classified according to the self-declared life satisfaction levels.

Starting from this preliminary analysis, this study wants to analyse the association between cultural participation activities and life satisfaction, exploring how this relationship is moderated by family background. However, before implementing more sophisticated models presented in Chap. 3, the large array of cultural activities included in the study engenders two main methodological challenges that must be addressed. First, there is high collinearity between the variables, especially regarding passive activities, as shown in Fig. 2. Second, it must be considered that people's cultural profile is also strongly characterised by socio-economic factors. To address both methodological challenges, we grouped individuals based on distinct patterns of cultural participation using Latent Profile Analysis (LPA), a clustering technique that has already been proven effective in capturing cultural profiles in a population (Mak et al., 2020). While other methods such as Factor Analysis can also be employed (Lee et al., 2021), LPA can also provide individuals' probabilities of belonging to each cultural profile.¹ Overall, this approach has two main advantages. First, it is able to capture the inner interconnections that exist among the engagement in different artistic forms. The omnivore theory posits that cultural tastes span across a spectrum of cultural activities, showing how the participation in each artistic form is not independent, but is all part of broader behavioural patterns (De Vries & Reeves, 2022). Moreover, methodologically, identifying synthetic cultural profiles limits the endogeneity issues related to overfitting and regressors' multicollinearity that would emerge when including all activities separately in the model.

Cultural activities are thus clustered in some groups based on the observed propensity to be consumed jointly and the intensity of their joint consumption, which is described over three levels for active participation, and four levels for passive activities, as detailed in Sect. 2. Moreover, to better identify the cultural consumption profiles, we also include information on people's socio-economic characteristics, expressed through their educa-

¹ For a robustness check, LPA results were compared with Factor Analysis scores, showing high positive correlations coefficients.

tional attainment and income range. These latter variables are indeed significant in shaping and explaining people's cultural engagement (Suarez-Fernandez et al., 2020). To standardize the values of the different variables, all metrics are scaled and centred at zero. This two-step approach preserves substantive interpretability as the estimated cultural profiles are theoretically grounded constructs, directly connected to the omnivore theory literature (Peterson, 1992).

Through LPA, several solutions are run and compared with Bayesian Information Criterion (BIC), going from a minimum of 2 to a maximum of 6 profiles (solutions with 7 clusters or more were discarded since the algorithm started to generate null values). The resulting BIC for each solution is presented in Table 4, and six profiles are selected as the best fit.

Figure 3 shows, for each of the six profiles, the average values of participation in each cultural activity and socio-economic characteristics. Considering this outcome, each cultural participation profile is now described, discussed, and named.

- Profile 1 represents the “*literary-artistic*” cultural participation, as it focuses more intensively on activities such as writing, reading, attending libraries, drawing, and theatre and museum attendance. These are individuals who can be considered highly oriented toward highbrow culture, and cultural activities with the highest level of symbolic meaning.
- Profile 2 has high socio-economic characteristics and average participation rates, especially regarding popular activities such as reading, museums, or taking photos. We named this cluster “*traditional*” cultural participation, since it is in fact mostly oriented toward more traditional and non-particularly dedicated cultural lifestyles.
- Profile 3 describes “*performing arts*” enthusiasts, as it consists of individuals with high participation rates in activities such as acting and dance, as well as events related to concerts and cinema, although not necessarily related to highbrow culture.
- Profile 4, instead, defines the “*music-lover*”, since it shows a specific dedication to music-related activities such as playing instruments, singing, cinema and concerts.
- Profile 5 describes the “*passive*” cultural participation, since it has an above-average dedication to all passive cultural activities: reading, cinemas, concerts, libraries, museums and theatres. It is also the group with the highest average income, confirming the strong link between passive participation and social capital already theorised by Bourdieu (1984), but also confirmed in more recent studies (Klokke & Jæger, 2022; Mak et al., 2021; Hanquinet et al., 2019).
- Profile 6 is made of “*non-participants*” since it is by far the least active group, with the lowest value of cultural participation both in a passive and active form. Not surprisingly, people belonging to that population group appear to have, on average, the lowest education and income.

Table 4 LPA model selection criteria

# Classes	logLIK	AIC	BIC	ICL
2	−26,624	53,308	53,546	−53,636
3	−26,369	52,904	53,224	−53,470
4	−25,656	52,207	52,609	−52,885
5	−25,429	51,676	52,161	−52,477
6	−24,720	49,645	50,213	−50,490

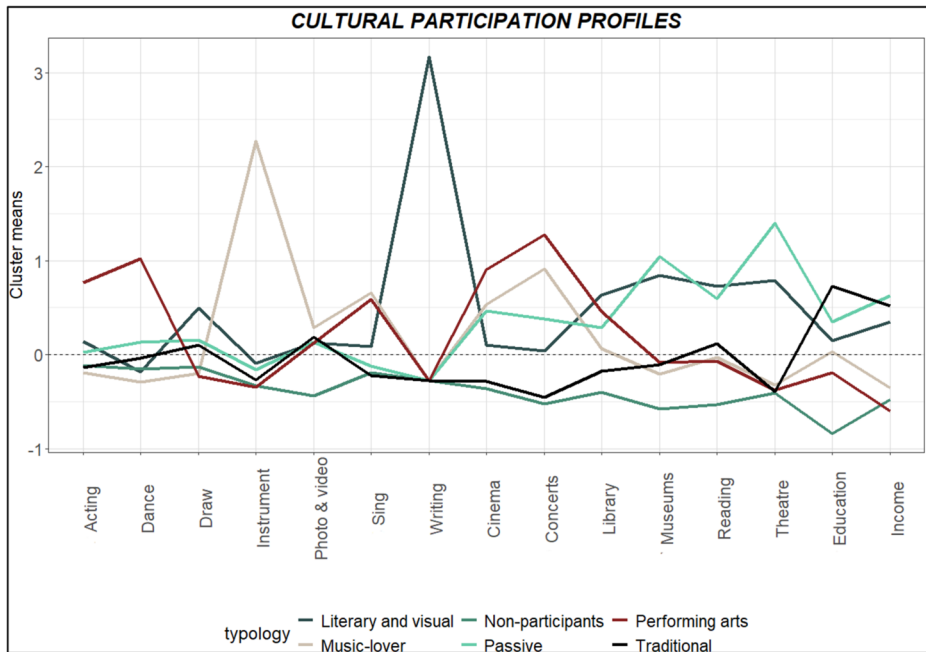


Fig. 3 Definition of cultural participation profiles

Individuals' posterior probabilities of belonging to these cultural participation profiles are used as continuous regressors rather than assigning each respondent to a single profile via hard classification. This preserves the full distributional information from the LPA and avoids the sharp misclassification errors that arise from deterministic assignment (Bolck et al., 2004).

The size of each profile is shown in Table 5. Once those cultural profiles are defined, the cultural participation of each respondent is expressed through the probability of belonging to each cluster.

Considering these groups, we now explore the relationship between people's cultural participation and life satisfaction, as well as the moderating role of family background, using FMM. As mentioned previously, when assessing the relationship with life satisfaction we assume that the population heterogeneity cannot be properly captured by a single equation. In other words, the link with cultural participation is not uniform across the whole population, but it is, instead, shaped by individual attitudes and behaviours. The full relationship is, instead, better explained by the existence of several sub-groups, each one with its own behavioural model.

By estimating the model with different numbers of population groups or latent classes, we use the Bayesian Information Criterion (BIC) associated with each FMM to determine the optimal number of classes in the population. Table 6 presents three information criteria for the FMM model selection. The AIC values decrease monotonically with the number of classes, yet the improvement diminishes substantially beyond two classes. The BIC and the entropy-based ICL, which penalises class overlap more heavily, favour the two-class solution. Given the convergence between BIC and ICL, and given the priority of parsimony and interpretability in a sample of this size, the two-class solution is retained as the reference

Table 5 Overview of the cultural participation typologies identified

Profile <i>n.</i>	Cultural part. typology	Size
1	<i>Literary-artistic</i>	93
2	<i>Traditional</i>	246
3	<i>Performing arts</i>	115
4	<i>Music-lover</i>	137
5	<i>Passive</i>	210
6	<i>Non-participants</i>	357
	<i>Total sample</i>	1,156

Table 6 FMM model selection criteria

# Classes	logLIK	AIC	BIC	ICL
2	−3280	6634	6786	7303
3	−3233	6585	6855	7599
4	−3209	6543	6970	7707

model throughout the analysis. Additionally, results are robust to the choice of two versus three classes (see Appendix A).

As a result, the two-group FMM is considered the reference empirical model. Figure 4 displays the density of life satisfaction for both latent classes, with Class 1 having a higher level of life satisfaction than Class 2. The first group is the smallest, with 550 respondents, while 608 respondents constitute the second group. Figure 4 also compares distributions of life satisfaction of the two groups identified with the life satisfaction of the full sample, which would be addressed by traditional models such as OLS (on the left).

Table 7 presents the results of the classification equation for the FMM with two classes. The estimated coefficients for the concomitant factors —i.e., the variables used to classify individuals into the two unobserved groups— determine the a priori probability of belonging to Class 2. Therefore, positive coefficients indicate a higher probability of being in Class 2, while negative coefficients suggest a greater likelihood of belonging to Class (1). For example, neither religious faith nor gender significantly affects the a priori probability of class membership. However, belonging to a non-disadvantaged family, being older, or being a student increases the likelihood of being part of the happiest group (Class 1). In contrast, younger people or those with more years of education have a higher probability of being classified into Class (2). This trend likely reflects the rising levels of education over time, as discussed in Sect. 2, and the tendency for income to increase with age. This outcome confirms the existing role of family in affecting and explaining people's levels of life satisfaction (Schnitzlein & Wunder, 2016; Headey et al., 2014).

To further differentiate the two classes, Table 8 presents the average means of each cultural profile being part of either Class 1 or Class 2. It also compares the average values of socio-demographic variables for both classes, with the last column reporting the p-value for testing the null hypothesis of equal means across the classes.² In terms of cultural participation, Class

² The discrepancies between the results of the classification equation presented in Table 7 and those in Table 8 for the same variables are due to the different natures of these results. The coefficients displayed in Table 7 are estimated under the ceteris paribus clause and define the a priori probability of belonging to one of the two classes. In contrast, the means presented in Table 8 are obtained after individuals are probabilistically assigned to the two classes of the FMM model, using posterior probabilities calculated based not only on the a priori probabilities but also on all the information used in the FMM model's estimation. Therefore, such differences may arise in some cases.

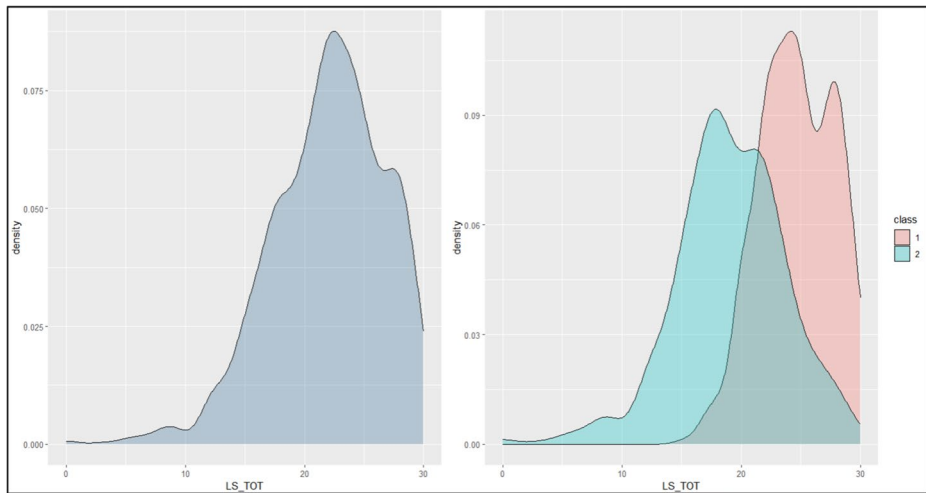


Fig. 4 Life satisfaction distribution for the whole sample and by latent class

Table 7 Classification equation of the FMM with two classes

Concomitant:	Coeff.	Stand. Dev.
Non-disadv. Family	-0.829*	(0.330)
Education (log)	1.124**	(0.404)
Age (log)	-2.190***	(0.592)
Male	-0.219	(0.278)
City population (log)	0.008	(0.081)
Atheist	0.626	(0.335)
Student	-2.771*	(1.110)
Unemployed	-0.222	(0.719)
Employed	-0.560	(0.468)
Region_north-east	1.050*	(0.428)
Region_north-west	-0.085	(0.379)
Region_south	0.749*	(0.347)

** $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

1 shows a higher incidence of individuals engaging in *literary-artistic* activities, that is, the most exclusive and highbrow activities, whereas Class 2 features more *traditional* participants and people involved in *performing arts*. Socio-economically, members of Class 1 are older individuals, who also tend to come from more advantaged families, though they have comparatively lower educational attainment, probably due to being from older, albeit relatively less well-educated, cohorts. This result is also supported by a lower incidence of sport activity - usually undertaken by younger cohorts - and by the higher proportion of people coming from the rich North-Western region. Moreover, despite gender not directly affecting the a priori probability of belonging to one of the classes, as shown in Table 7, Class 1 has a higher prevalence of males. In contrast, members of Class 2, tend to be younger, with higher levels of education, have more people who declare themselves as atheists and come from less advantaged family backgrounds, with a higher probability of being unemployed and from disadvantaged regions, especially the South of Italy. These findings confirm the indirect asso-

Table 8 Mean and Standard deviation (in parentheses) by Latent Class

	Class 1 (high life satisfaction)		Class 2 (low life satisfaction)		P-value*
Live satisfaction	25.067	(2.881)	19.443	(4.537)	<0.001
Literary-artistic	0.117	(0.322)	0.050	(0.219)	<0.001
Traditional	0.187	(0.334)	0.242	(0.339)	0.007
Performing arts	0.071	(0.220)	0.130	(0.296)	<0.001
Music-lover	0.096	(0.278)	0.125	(0.309)	0.222
Passive	0.178	(0.351)	0.176	(0.346)	0.934
Gender_male	0.508	(0.500)	0.440	(0.498)	0.012
Age (log)	3.899	(0.401)	3.633	(0.371)	<0.001
Non-disadv. Family	0.860	(0.347)	0.626	(0.484)	<0.001
Education years (log)	2.457	(0.440)	2.644	(0.294)	<0.001
Income (log)	7.735	(4.211)	7.945	(3.938)	0.381
Sport_yes	0.590	(0.492)	0.731	(0.444)	<0.001
Atheist	0.182	(0.386)	0.307	(0.461)	<0.001
Student	0.126	(0.333)	0.047	(0.212)	<0.001
Unemployed	0.110	(0.326)	0.156	(0.349)	0.021
Employed	0.506	(0.500)	0.715	(0.451)	<0.001
Region_north_east	0.136	(0.343)	0.264	(0.441)	<0.001
Region_north_west	0.345	(0.476)	0.203	(0.402)	<0.001
Region_south	0.262	(0.440)	0.373	(0.484)	<0.001
N	550		608		

* P-value of the corresponding t-test on mean differences

ciation between family background, and education in general, and overall life satisfaction, as individuals with better-educated parents are more likely to belong to the happiest group.

After describing the two resulting groups of people, Table 9 presents the estimated parameters of the two behavioural models linking cultural participation with life satisfaction. To compare these results with those of the traditional model, the last column reports the outcomes of the OLS estimation, which applies the same analysis to the entire sample, assuming no differences in the relationship between life satisfaction and cultural participation across the population. As expected, the OLS coefficients primarily reflect an average result between Class 1 and Class 2 for each independent variable. This occurs because the OLS treats the entire sample as a single population group, thereby assuming less within-sample heterogeneity than in the class-specific analysis.

These behavioural functions, as discussed earlier, include variables defining cultural consumption profiles, along with sport activity and personal income as indicators of people's hedonic purchases. It is important to underline that FMM identifies classes based on both the baseline levels of life satisfaction and the distinct mechanisms through which individuals derive utility from cultural participation. The separation between Class 1 (higher baseline LS) and Class 2 (lower baseline LS) reflects both how much satisfaction individuals experience overall and how they achieve that satisfaction through cultural activities. First, the constant value represents, for each class separately, the baseline value of life satisfaction. While Class 1 has a baseline value of 22.8, Class 2's baseline is 13.6 points, certifying the large existing gap between the two classes, as discussed in the previous paragraphs. Beyond this distinction, notable differences in the significance and magnitude of the relationships between covariates and life satisfaction emerge. For Class 2 (lower baseline LS),

Table 9 Mean and Standard deviation (in parentheses) of FMM and OLS coefficients

	FMM		OLS
	Class 1 (high LS)	Class 2 (low LS)	
<i>Literary-artistic</i>	4.292*** (0.581)	2.655 (1.963)	4.264*** (0.556)
<i>Traditional</i>	3.366*** (0.657)	4.417*** (0.867)	3.220 *** (0.364)
<i>Performing arts</i>	1.679 (0.891)	2.705*** (0.861)	1.488** (0.587)
<i>Music-lover</i>	0.264 (1.055)	4.600*** (0.864)	2.321*** (0.527)
<i>Passive</i>	3.362*** (0.612)	5.418*** (0.807)	4.260*** (0.482)
<i>Sport</i>	0.006 (0.376)	1.478 (0.780)	0.043 (0.320)
<i>Income (log)</i>	-0.040 (0.050)	0.244*** (0.075)	0.103*** (0.036)
<i>Constant</i>	22.843*** (0.678)	13.626*** (0.913)	19.931*** (0.364)
N	550	608	1,158

* P -value * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

most of the cultural profiles are positively correlated with life satisfaction, particularly *traditional* (+4.4), *music-lover* (+4.6) and *passive* participants (+5.4), while *performing arts* have a smaller coefficient (+2.7). The only cultural profile not significantly associated with life satisfaction for Class 2 is the *literary-artistic* cluster. This result may be explained by the lower cultural capital of people in this class, who primarily come from disadvantaged families. The high symbolic value attached to the *literary-artistic* cluster often requires adequate intellectual and conceptual tools for full appreciation, suggesting a potential link between family background and the ability to derive utility from this type of cultural activity.

Thus, the results emphasize the central role of parents' human capital in developing children's cultural capital. Furthermore, this finding also implies the presence of a tangible barrier for individuals from disadvantaged backgrounds who, on one hand, are less likely to engage in such activities (as shown in Table 2, especially regarding writing), and, on the other hand, may struggle to translate their engagement in these highbrow cultural activities into life satisfaction as effectively as their more privileged counterparts. Besides cultural activities, both *sports activities* (p -value < 0.1) and *personal income* are positively correlated with life satisfaction for this Class. This is consistent with the expected link between income and well-being, highlighting the positive role of hedonic purchases in enhancing life satisfaction in this class (Class 2).

In contrast, the behavioural model for the happier Class 1 (higher baseline LS) shows some differences. First, while *traditional*, *passive* and *performing arts* (p -value < 0.1) profiles are again positively associated with life satisfaction, the coefficients are smaller (+3.4, and +1.7, respectively). However, the *literary-artistic* profile shows the largest significant link (+4.3), much larger than in Class 2, where it is not significant. This reinforces the earlier discussion about the symbolic meaning of these forms of cultural expression, which appear to be more predominantly correlated in non-disadvantaged segments of the population. This cultural pro-

file is also associated with higher socio-economic status and is more prevalent in Class 1, suggesting a distinctive social dimension to this form of cultural engagement. In this regard, parents' social capital plays an important role in acquiring bonds and connections (Molina et al., 2011), shaping, in turn, the link between these factors and life satisfaction.

Finally, *music-lover* participation is not significant for Class 1, in contrast to the strong correlation that emerged for Class 2. This profile includes popular activities such as cinema, concerts, instrument playing, and singing—many of which are practiced regardless of family background (as shown in Table 2). The greater coefficient of music for Class 2 could relate to its comforting nature, often cited as a powerful tool against sadness, especially among underprivileged groups (Gómez-Zapata et al., 2021). As for the other variables in the behavioural equation, neither sports activity nor personal income significantly affects life satisfaction in Class 1. This aligns with existing research showing that the relationship between income and well-being is stronger among poorer segments of the population and tends to weaken among wealthier groups (Hajek & König, 2019).

Overall, these results are in line with existing studies that highlighted the positive relationship between of cultural participation and life satisfaction, since all profiles are significant for at least one class (Baldin & Bille, 2023; Reyes-Martínez et al., 2021). Conversely, the heterogeneity of the association between cultural participation typologies and life satisfaction across the two identified classes may help explain why some studies observed a weak or even non-significant link (Węziak-Białowolska, 2016; Michalos & Kahlke, 2008). This result suggests the existence of different sub-groups in the population who exhibit distinct attitudes and characteristics, and therefore experience a different relationship between cultural behaviours and life satisfaction. In this setting, while Class 1 already enjoys higher levels of life satisfaction and only a few activities (mainly those related to social interaction) are significant, cultural participation represents an important factor primarily for poorer and more disadvantaged parts of the population belonging to Class 2, helping to mitigate the existing inequalities in terms of life satisfaction between the two groups.

Finally, regarding the role of family background, it emerges as an indirect factor in the acquisition of life satisfaction. More precisely, it plays a significant role in affecting people's propensity to belong to the happier subgroup of the population, which in turn determines the behaviours through which they extract life satisfaction from cultural participation. This latter distinction, indirectly explained by family background, is particularly evident in activities associated with higher socio-economic status, such as literary, heritage-related and visual arts. Individuals from more advantaged families are able to engage more effectively in these activities, thereby demonstrating stronger associations with life satisfaction. This suggests that the moderating role of family background may be attributed to the social connections acquired through parents. This hypothesis aligns with Bourdieu's (1984) theories, which argue that culture acts as a mechanism of social distinction, primarily serving the most privileged segments of society. Taken together, the mechanisms explaining people's overall life satisfaction are thus transmitted across generations and may contribute to generating structural inequalities of opportunity in well-being, as measured for instance by Schnitzlein and Wunder (2016) and Headley et al. (2014). This result highlights the importance of formal and non-formal processes of education led by the family, which could affect the possibility of benefiting from cultural engagement.

As a robustness check, the FMM was estimated with three latent classes. The results, reported in Appendix A, are broadly consistent with the two-class solution. The three-class

model disaggregates the lower-life-satisfaction group into two subgroups — one with a low baseline and one with an intermediate baseline — while the high-life-satisfaction class remains structurally similar to Class 1 in the main model. The direction, magnitude, and significance of the cultural profile coefficients are largely preserved across the three-class solution. Family background and education remain the dominant concomitant factors across all classes, and the pattern by which the literary-artistic profile is less significantly associated with life satisfaction for the least satisfied group persists. The main substantive findings are therefore not sensitive to the choice of two versus three classes, providing additional confidence in the conclusions drawn from the reference model.

5 Conclusions

While investigating the association between cultural participation and life satisfaction, the study examines how this relationship varies across different sub-groups of people and highlights the moderating role of family background. Two main outcomes of this research can be identified. First, cultural participation appears to be positively associated with life satisfaction, particularly among poorer individuals, those from less privileged families and least happy segments of the population, as well as younger individuals. In fact, all the cultural participation profiles identified are strongly associated with life satisfaction for this under-privileged group, suggesting its importance for policy as a means to reduce life satisfaction disparities. However, second, the use of FMM suggests the existence of different behavioural patterns within the population. These results indicate that the link between people's life satisfaction and their actions, including their cultural engagement, is not univocal, but changes across subgroups. Among the most significant factors, family background emerges as a crucial determinant of this heterogeneity. In fact, in this study, family background relates to the probability of belonging to the happier class and, most of all, the ability to grasp higher utility from engagement in certain cultural activities. This is particularly visible for the *literary-artistic* engagement profile, which is positively associated with life satisfaction only for people coming from educated families.

As a consequence, while the promotion of cultural participation could play a relevant role for enhancing well-being, it may not be sufficient to address the most disadvantaged parts of the population and the existing social inequalities. Policies promoting cultural participation should be accompanied by complementary investments in cultural literacy and family-level educational support to ensure equitable benefits. This result supports the development of more advanced well-being public policy theories that are currently still deficient (Fabian et al., 2024).

Alongside these encouraging results, several limitations and opportunities for further research can be identified. Firstly, results do not investigate any causal impact, but rather focus on the overall association existing between cultural participation and well-being. Also, findings are specifically tied to the Italian context, which could have affected the cultural profiles identified, and the relevance of family background. Future studies could replicate this analysis in different countries, comparing the results and investigating whether the relevance of cultural participation for disadvantaged people remains consistent. Considering cultural participation, we recognize that the two-step approach may not fully propagate the uncertainty from the LPA into the FMM estimation, leading to potential classification errors. Future work

could explore joint modeling approaches to address this concern, while still accounting for the existence of latent patterns of cultural engagement. The validity of results could also be tested using a more diverse array of metrics related to well-being, such as perceived health status, to provide a more rounded picture of the relationship highlighted. Moreover, future studies could explore the same analysis using non-parametric or semi-parametric approaches to grasp non-linear relationships between the variables. Finally, the most disadvantaged populations were only partially considered in our study due to the difficulty of reaching these individuals through internet-based surveys, leading to their under-representation in responses. Future research should address this limitation by conducting targeted investigations to gather more robust evidence on the nature and significance of the relationship being studied.

Appendix 1 Sensitivity Analysis

To validate the robustness of the main conclusions of the study, a sensitivity analysis is run, setting three classes in the FMM. As a result, three main sub-groups are identified in the population, each one with its own behavioural model. Figure 5 shows the life satisfaction distribution across the three classes obtained.

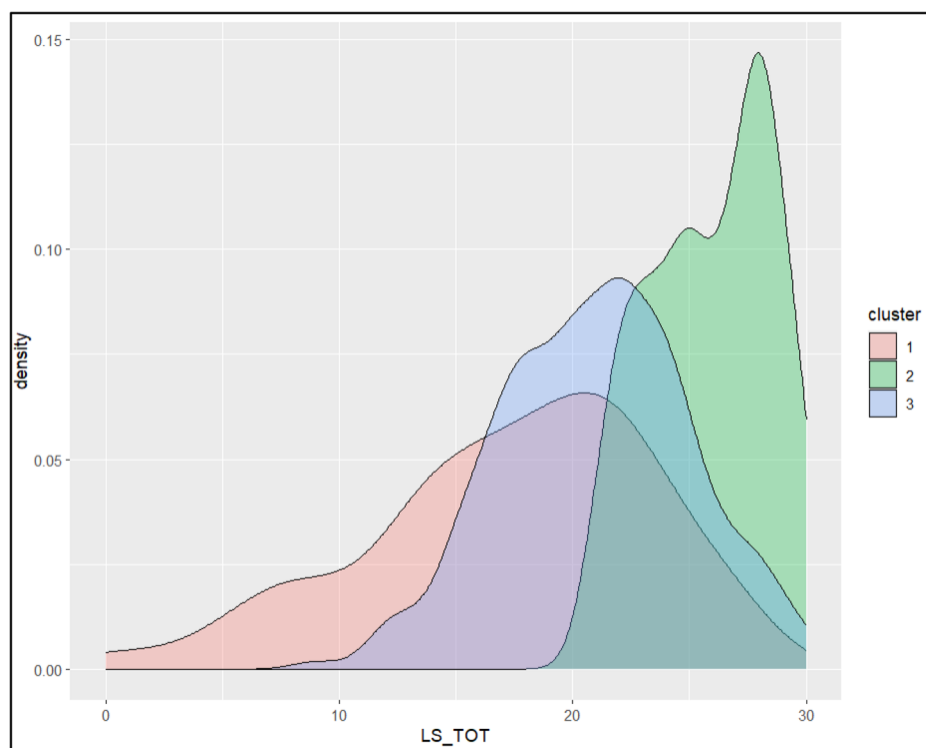


Fig. 5 Life satisfaction distribution for the three latent classes

In this new setting, Class 1 is the smallest one (180 individuals), and it captures people with the lowest self-declared life satisfaction levels. In contrast, Class 2 is the most numerous class (708 observations) and it is composed of people with the highest life satisfaction levels. Finally, Class 3 is composed of 330 people with in-between levels of life satisfaction with respect to the other two classes. To characterise the Classes further, Table 10 presents the concomitants affecting the probability of pertaining to each of them. In line with the main results, family background emerges as a significant determinant, as people coming from more disadvantaged families is more likely to pertain in the class with lower life satisfaction. Moreover, this latter class is composed of younger people or those with more years of education.

Similarly, to the main analysis with two classes, Table 11 presents the estimated parameters of the three behavioural models with the determinants of life satisfaction for each class. First, the baselines reflect the specification discussed previously. Classes can be ordered in life satisfaction terms: Class 1 is the least happy (baseline of 11.3), then Class 3 (baseline of 15.0), and finally Class 2 (baseline of 23.7). Considering the correlation between cultural participation and life satisfaction, several interesting insights emerge from this specification, confirming and extending the results that emerged in the main analysis of this study regarding the diverging behavioural patterns of utility acquisition. For Class 3, with the average life satisfaction levels, all cultural profiles are significantly and positively associated with life satisfaction. Similarly, also in Class 2 cultural profiles show a positive association, except for music-lovers. Instead, Class 1, with the lowest life satisfaction baseline and characterised by people coming from more disadvantaged families, the link between life satisfaction and cultural participation is less clear. This finding suggests the potential existence of barriers hampering the association between culture and life satisfaction in people from disadvantaged backgrounds

Table 10 Classification equation of the FMM with three classes

Concomitant:	Class 2 (high LS)		Class 3 (average LS)	
	Coeff.	Stand. Dev.	Coeff.	Stand. Dev.
Non-disadv. Family	1.367*	(0.556)	1.209*	(0.612)
Education (log)	-3.883***	(1.015)	-3.524**	(1.092)
Age (log)	0.657	(1.062)	-2.542*	(1.234)
Male	0.872	(0.532)	0.668	(0.565)
City population (log)	0.225	(0.170)	0.354	(0.184)
Atheist	-1.373*	(0.559)	-1.248*	(0.601)
Student	-3.552	(1.956)	-14.153	(10.220)
Unemployed	-2.320	(1.384)	-4.121**	(1.551)
Employed	-0.339	(1.069)	-1.384	(0.612)
Region_north-east	-0.790	(0.872)	0.489	(0.955)
Region_north-west	0.896	(0.779)	1.079	(0.868)
Region_south	-1.123	(0.612)	-0.396	(0.677)

* P -value * $p < 0.05$, ** $p < 0.01$,
*** $p < 0.001$

Table 11 Mean and Standard deviation (in parentheses) of FMM with three classes

	Class 1 (low LS)	Class 2 (high LS)	Class 3 (average LS)
<i>Literary-artistic</i>	0.254 (2.230)	4.019*** (0.590)	5.325*** (0.859)
<i>Traditional</i>	8.260** (2.744)	3.398*** (0.715)	4.414*** (0.777)
<i>Performing arts</i>	3.835 (1.974)	3.717*** (0.985)	2.131** (0.761)
<i>Music-lover</i>	4.085* (1.911)	-0.933 (0.872)	4.769*** (0.730)
<i>Passive</i>	4.981 (2.631)	2.771*** (0.621)	5.396*** (0.660)
<i>Sport</i>	5.659*** (1.675)	0.117 (0.375)	-0.125 (0.437)
<i>Income (log)</i>	-0.474* (0.220)	0.006 (0.045)	0.329*** (0.075)
<i>Constant</i>	11.346*** (2.306)	23.734*** (0.399)	15.003*** (0.796)
N	120	330	708

* P -value * $p < 0.05$, ** $p < 0.01$,
*** $p < 0.001$

Funding Open access funding provided by Politecnico di Milano within the CRUI-CARE Agreement.

Data Availability The data that support the findings of this study are available from Digital Innovation for Culture Observatory of Politecnico di Milano (<https://eng.osservatori.net/digital-innovation-in-culture-eng/>), but restrictions apply to the availability of these data, which were used under licence for the current study and so are not publicly available. The data are, however, available from the authors upon reasonable request and with the permission of the Digital Innovation for Culture Observatory.

Declarations

Disclosure The authors have no relevant financial or non-financial interests to disclose. This work was funded by the grant of the Ministerio de Ciencia e Innovacion del Gobierno de Espana PID2022-137352NB-C43. Juan Prieto-Rodriguez is members of the “CULTURALITY; CULTUral heritage in RurAL remote areas for creative tourism and sustainability” project, funded by the European Commission (HORIZON-CL2-2023-HERITAGE-0110113262).

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article’s Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article’s Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Aitkin, M., & Rubin, D. B. (1985). Estimation and hypothesis testing in finite mixture models. *Journal of the Royal Statistical Society. Series B, Statistical Methodology*, 47(1), 67–75. <https://doi.org/10.1111/j.2517-6161.1985.tb01331.x>

- Arcidiacono, P., & Jones, J. B. (2003). Finite mixture distributions, sequential likelihood and the EM algorithm. *Econometrica*, 71(3), 933–946. <https://doi.org/10.1111/1468-0262.00431>
- Ateca-Amestoy, V. (2008). Determining heterogeneous behavior for theater attendance. *Journal of Cultural Economics*, 32(2), 127–151. <https://doi.org/10.1007/s10824-008-9065-z>
- Ateca-Amestoy, V., & Ugidos, A. (2013). The Impact of Different Types of Resource Transfers on Individual Wellbeing: An Analysis of Quality of Life Using CASP-12. *Social Indicators Research*, 110(3), 973–991. <https://doi.org/10.1007/s11205-011-9968-5>
- Báez-Montenegro, A., Prieto-Rodríguez, J., & Salas, R. (2011). Chapter 7 Changes Over Time in Parental Education and Inequality of Opportunities in Chile. In J. G. Rodríguez (Ed.), *Research on Economic Inequality* (Vol. Vol. 19, pp. 157–177). Emerald Group Publishing Limited. [https://doi.org/10.1108/S1049-2585\(2011\)0000019010](https://doi.org/10.1108/S1049-2585(2011)0000019010)
- Baldin, A., & Bille, T. (2023). The lost value for users of cultural institutions during the COVID-19 pandemic: A life satisfaction approach. *International Review of Economics*, 70(2), 257–281. <https://doi.org/10.1007/s12232-023-00418-7>
- Blanchflower, D. G., & Clark, A. E. (2021). Children, unhappiness and family finances. *Journal of Population Economics*, 34(2), 625–653. <https://doi.org/10.1007/s00148-020-00798-y>
- Bolck, A., Croon, M., & Hagenaars, J. (2004). Estimating Latent Structure Models with Categorical Variables: One-Step Versus Three-Step Estimators. *Political Analysis*, 12(1), 3–27. <https://doi.org/10.1093/pan/12mp001>
- Bourdieu, P. (1984). *Distinction: A Social Critique of the Judgement of Taste*. https://www.academia.edu/48726680/Distinction_A_Social_Critique_of_the_Judgement_of_Taste_Pierre_Bourdieu
- Bourguignon, F., Ferreira, F. H. G., & Menéndez, M. (2007). Inequality of Opportunity in Brazil. *Review of Income and Wealth*, 53(4), 585–618. <https://doi.org/10.1111/j.1475-4991.2007.00247.x>
- Brown, J. L., MacDonald, R., & Mitchell, R. (2015). Are people who participate in cultural activities more satisfied with life? *Social Indicators Research*, 122(1), 135–146. <https://doi.org/10.1007/s11205-014-0678-7>
- Cerisola, S., & Panzera, E. (2022). Cultural participation in cultural and creative cities: Positive regional outcomes and potential congestion concerns. *Papers in Regional Science*, 101(6), 1245–1261. <https://doi.org/10.1111/pirs.12709>
- Checchi, D., & Peragine, V. (2010). Inequality of opportunity in Italy. *The Journal of Economic Inequality*, 8(4), 429–450. <https://doi.org/10.1007/s10888-009-9118-3>
- De Vries, R., & Reeves, A. (2022). What does it mean to be a cultural omnivore? Conflicting visions of omnivorousness in empirical research. *Sociological Research Online*, 27(2), 292–312. <https://doi.org/10.1177/13607804211006109>
- Diener, E. (Ed.). (2009). *Assessing Well-Being* (Vol. 39). Springer Netherlands. <https://doi.org/10.1007/978-90-481-2354-4>
- European Union (2007). *Treaty of Lisbon. Official Journal of the European Union*. https://ec.europa.eu/archives/lisbon_treaty/full_text/
- Fabian, M., Alexandrova, A., Coyle, D., Agarwala, M., & Felici, M. (2024). Wellbeing public policy needs more theory. *International Journal of Wellbeing*, 14(3), 1–17. <https://doi.org/10.5502/ijw.v14i3.2691>
- Fancourt, D., & Finn, S. (2019). *What is the evidence on the role of the arts in improving health and well-being? A scoping review*. WHO Regional Office for Europe. <http://www.ncbi.nlm.nih.gov/books/NBK553773/>
- Fancourt, D., & Steptoe, A. (2019). Cultural engagement and mental health: Does socio-economic status explain the association? *Social Science & Medicine*, 236, Article 112425. <https://doi.org/10.1016/j.socscimed.2019.112425>
- Fancourt, D., & Warran, K. (2024). a fRAMework of the Determinants of Arts and Cultural Engagement (RADIANCE): Integrated insights from ecological, behavioural and complex adaptive systems theories. *Wellcome Open Research*, 9, 356. <https://doi.org/10.12688/wellcomeopenres.21625.1>
- Fancourt, D., Stringaris, A., & Sacco, P. L. (2026). Mechanisms underpinning the mental health impact of arts engagement. *Nature Reviews Psychology*, 5(4), 290–302. <https://doi.org/10.1038/s44159-026-00545-2>
- Fernandez-Blanco, V., Orea, L., & Prieto-Rodríguez, J. (2009). Analyzing consumers heterogeneity and self-reported tastes: An approach consistent with the consumer's decision making process. *Journal of Economic Psychology*, 30(4), 622–633. <https://doi.org/10.1016/j.joep.2009.04.005>
- Gladstone, J. J., Ruberton, P. M., Margolis, S., & Lyubomirsky, S. (2024). Does variety in hedonic spending improve happiness? Testing alternative causal mechanisms between hedonic variety and subjective well-being. *BMC Psychology*. <https://doi.org/10.1186/s40359-024-01599-8>
- Gómez-Zapata, J. D., Herrero-Prieto, L. C., & Rodríguez-Prado, B. (2021). Does music soothe the soul? Evaluating the impact of a music education programme in Medellín, Colombia. *Journal of Cultural Economics*, 45(1), 63–104. <https://doi.org/10.1007/s10824-020-09387-z>

- Grossi, E., Tavano Blessi, G., Sacco, P. L., & Buscema, M. (2012). The interaction between culture, health and psychological well-being: Data mining from the Italian culture and well-being project. *Journal of Happiness Studies*, 13(1), 129–148. <https://doi.org/10.1007/s10902-011-9254-x>
- Hajek, A., & König, H. H. (2019). Not Getting What You Want? The Impact of Income Comparisons on Subjective Well-Being—Findings of a Population-Based Longitudinal Study in Germany. *International Journal of Environmental Research and Public Health*, 16(15), 2655. <https://doi.org/10.3390/ijerph16152655>
- Han, J., & Kaiser, C. (2024). Time use and happiness: US evidence across three decades. *Journal of Population Economics*, 37(1), 15. <https://doi.org/10.1007/s00148-024-00982-4>
- Hanquinet, L., O'Brien, D., & Taylor, M. (2019). The coming crisis of cultural engagement? Measurement, methods, and the nuances of niche activities. *Cultural Trends*, 28(2–3), 198–219. <https://doi.org/10.1080/09548963.2019.1617941>
- Headey, B., Muffels, R., & Wagner, G. G. (2014). Parents Transmit Happiness Along with Associated Values and Behaviors to Their Children: A Lifelong Happiness Dividend? *Social Indicators Research*, 116(3), 909–933. <https://doi.org/10.1007/s11205-013-0326-7>
- Iida, S. (2019). Structure of life satisfaction from the perspective of arts experiences in Japan. *Palgrave Communications*, 5(1), 143. <https://doi.org/10.1057/s41599-019-0354-2>
- Jensen, A., Folker, A. P., Lindström, M., & Ekholm, O. (2023). Arts and culture engagement for health: A Danish population-based study. *Public Health*, 225, 120–126. <https://doi.org/10.1016/j.puhe.2023.09.012>
- Kasahara, H., & Shimotsu, K. (2009). Nonparametric Identification of Finite Mixture Models of Dynamic Discrete Choices. *Econometrica*, 77(1), 135–175. <https://doi.org/10.3982/ECTA6763>
- Klokke, R. H., & Jäger, M. M. (2022). Family background and cultural lifestyles: Multigenerational associations. *Poetics*, 92, Article 101662. <https://doi.org/10.1016/j.poetic.2022.101662>
- Lee, C.-W., Lin, L.-C., & Hung, H.-C. (2021). Art and cultural participation and life satisfaction in adults: The role of physical health, mental health, and interpersonal relationships. *Frontiers in Public Health*, 8, Article 582342. <https://doi.org/10.3389/fpubh.2020.582342>
- Lindström, M., Pirouzifard, M., & Jensen, A. (2024). Arts, culture and sports engagement and self-rated health: A cross-sectional population-based study in southern Sweden. *BMC Public Health*, 24(1), Article 2654. <https://doi.org/10.1186/s12889-024-20031-9>
- Mak, H. W., Coulter, R., & Fancourt, D. (2020). Does arts and cultural engagement vary geographically? Evidence from the UK household longitudinal study. *Public Health*, 185, 119–126. <https://doi.org/10.1016/j.puhe.2020.04.029>
- Mak, H. W., Coulter, R., & Fancourt, D. (2021). Associations between neighbourhood deprivation and engagement in arts, culture and heritage: Evidence from two nationally-representative samples. *BMC Public Health*, 21(1), 1685. <https://doi.org/10.1186/s12889-021-11740-6>
- Masyn, K. E. (2013). *Latent Class Analysis and Finite Mixture Modeling*. Oxford University Press. <https://doi.org/10.1093/oxfordhb/9780199934898.013.0025>
- McCarthy, K., Ondaatje, E., Zakaras, L., & Brooks, A. (2004). *Gifts of the Muse: Reframing the Debate About the Benefits of the Arts*. RAND Corporation. <https://doi.org/10.7249/MG218>
- McLachlan, G. J., Lee, S. X., & Rathnayake, S. I. (2019). Finite mixture models. *Annual Review of Statistics and Its Application*, 6(1), 355–378. <https://doi.org/10.1146/annurev-statistics-031017-100325>
- Michalos, A. C., & Kahlke, P. M. (2008). Impact of arts-related activities on the perceived quality of life. *Social Indicators Research*, 89(2), 67.
- Molina, J. A., Navarro, M., & Walker, I. (2011). Intergenerational well-being mobility in Europe: Intergenerational well-being mobility in Europe. *Kyklos*, 64(2), 253–270. <https://doi.org/10.1111/j.1467-6435.2011.00505.x>
- Nononen, T., Kaikkonen, R., Murto, J., & Luoma, M. L. (2014). Cultural services and activities: The association with self-rated health and quality of life. *Arts & Health*, 6(3), 235–253. <https://doi.org/10.1080/17533015.2014.897958>
- OECD. (2013). *OECD Guidelines on Measuring Subjective Well-being*. OECD. <https://doi.org/10.1787/9789264191655-en>
- Pavot, W., & Diener, E. (1993). Review of the Satisfaction With Life Scale. In E. Diener (Ed.), *American Psychological Association* (Vol. 5, pp. 101–117). Springer Netherlands. https://doi.org/10.1007/978-90-481-2354-4_5
- Perez-Mayo, J. (2019). Inequality of opportunity, a matter of space? *Regional Science Policy & Practice*, 11(1), 71–87. <https://doi.org/10.1111/rsp3.12181>
- Pessoa, J., & Deloumeaux, L. (2009). *The 2009 Unesco framework for cultural statistics (FCS)*. UNESCO Institute for Statistics.
- Peterson, R. A. (1992). Understanding audience segmentation: From elite and mass to omnivore and univore. *Poetics*, 21(4), 243–258. [https://doi.org/10.1016/0304-422X\(92\)90008-Q](https://doi.org/10.1016/0304-422X(92)90008-Q)

- Peterson, R. A., & Kern, R. M. (1996). Changing Highbrow Taste: From Snob to Omnivore. *American Sociological Review*, 61(5), 900. <https://doi.org/10.2307/2096460>
- Prieto-Rodríguez, J., & Vecco, M. (2021). Reading between the lines in the art market: Lack of transparency and price heterogeneity as an indicator of multiple equilibria. *Economic Modelling*, 102, Article 105587. <https://doi.org/10.1016/j.econmod.2021.105587>
- Reeves, A., & de Vries, R. (2019). Can cultural consumption increase future earnings? Exploring the economic returns to cultural capital. *The British Journal of Sociology*, 70(1), 214–240. <https://doi.org/10.1111/1468-4446.12374>
- Reyes-Martínez, J. (2021). Cultural Participation and Subjective Well-Being in Latin America: Does Ethnic-Racial Ascription Matter? *Scientific Annals of Economics and Business*, 68, 53–70. <https://doi.org/10.47743/saeb-2021-0031>
- Reyes-Martínez, J., Takeuchi, D., Martínez-Martínez, O. A., & Lombe, M. (2021). The Role of Cultural Participation on Subjective Well-Being in Mexico. *Applied Research in Quality of Life*, 16(3), 1321–1341. <https://doi.org/10.1007/s11482-020-09811-8>
- Roemer, J. E. (1993). A Pragmatic Theory of Responsibility for the Egalitarian Planner. *Philosophy & Public Affairs*, 22(2), 146–166.
- Schnitzlein, D. D., & Wunder, C. (2016). Are we architects of our own happiness? The importance of family background for well-being. *The B.E. Journal of Economic Analysis & Policy*, 16(1), 125–149. <https://doi.org/10.1515/bejeap-2015-0037>
- Stevenson, D., Balling, G., & Kann-Rasmussen, N. (2017). Cultural participation in Europe: Shared problem or shared problematisation? *International Journal of Cultural Policy*, 23(1), 89–106. <https://doi.org/10.1080/10286632.2015.1043290>
- Suarez-Fernandez, S., Prieto-Rodríguez, J., & Perez-Villadoniga, M. J. (2020). The changing role of education as we move from popular to highbrow culture. *Journal of Cultural Economics*, 44(2), 189–212. <https://doi.org/10.1007/s10824-019-09355-2>
- UNESCO (2022). *UNESCO strategy on education for health and well-being*. <https://unesdoc.unesco.org/ark:/48223/pf0000381728>
- United Nations (2023). *United Nations System Mental Health and Well-being Strategy for 2024 and beyond*.
- van Wel, F., Couwenbergh-Soeterboek, N., Couwenbergh, C., ter Bogt, T., & Raaijmakers, Q. (2006). Ethnicity, youth cultural participation, and cultural reproduction in the Netherlands. *Poetics*, 34(1), 65–82. <https://doi.org/10.1016/j.poetic.2005.06.001>
- Vecco, M., Prieto-Rodríguez, J., & Teerink, S. (2024). Climbing the ladder? The gender gap in art prices across artists' cohorts in the Dutch art market. *European Economic Review*, 163, 104657. <https://doi.org/10.1016/j.euroecorev.2023.104657>
- Vegheš, C. (2020). Cultural Consumption as a Trait of a Sustainable Lifestyle: Evidence from the European Union. *European Journal of Sustainable Development*, 9(4), 125. <https://doi.org/10.14207/ejsd.2020.v9n4p125>
- Viola, E., Martorana, M., Ceriotti, D., De Vito, M., De Ambrosi, D., & Faggiano, F. (2024). The effects of cultural engagement on health and well-being: A systematic review. *Frontiers in Public Health*, 12, Article 1369066. <https://doi.org/10.3389/fpubh.2024.1369066>
- Węziak-Białowolska, D. (2016). Attendance of cultural events and involvement with the arts—Impact evaluation on health and well-being from a Swiss household panel survey. *Public Health*, 139, 161–169. <https://doi.org/10.1016/j.puhe.2016.06.028>